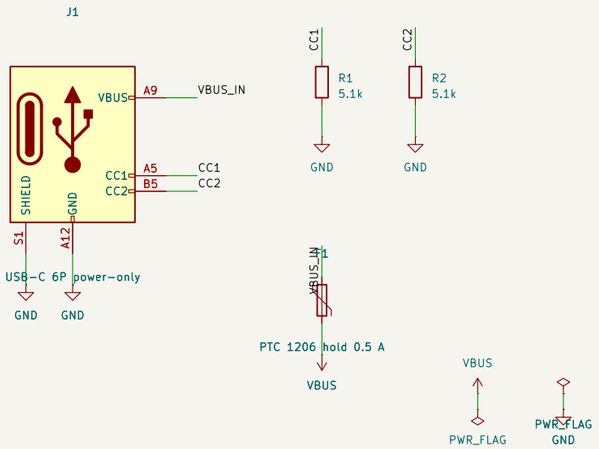
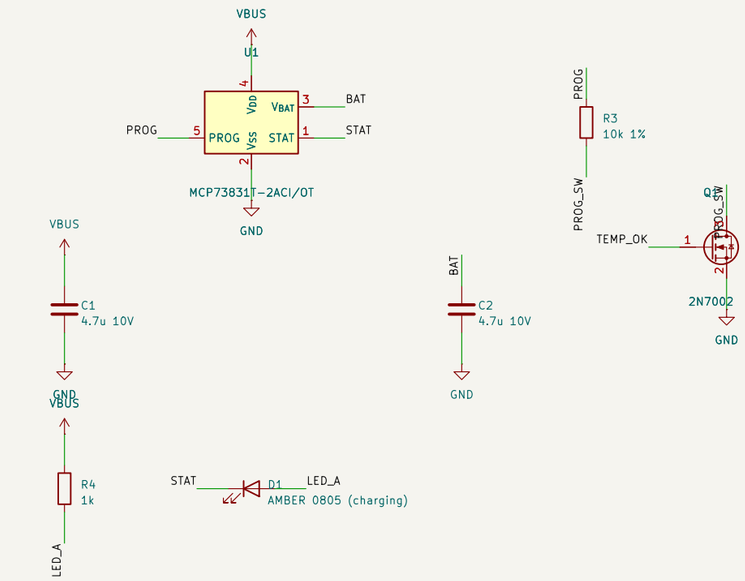
NIVA CHARGING DOCK — ENGINEERING PROTOTYPE — NOT FOR FABRICATION

01 USB-C SINK (5 V ONLY)

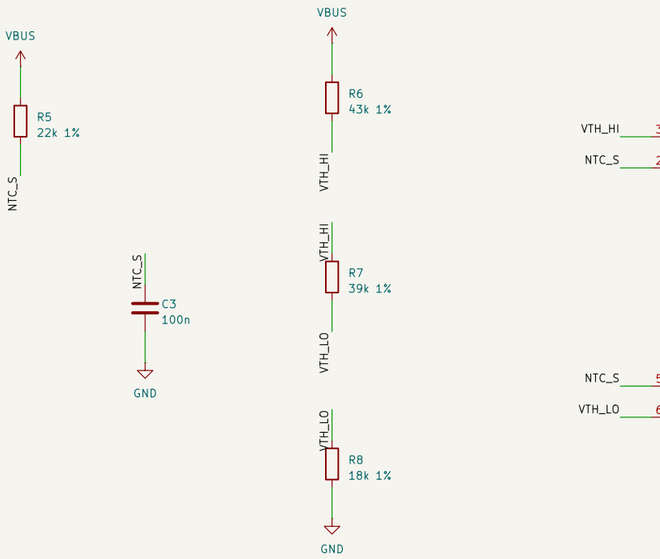


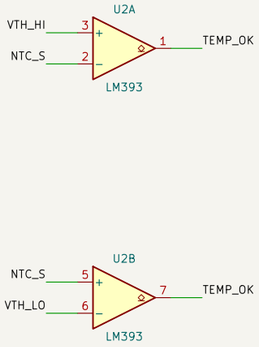
02 LINEAR CHARGER 4.20 V / 100 mA

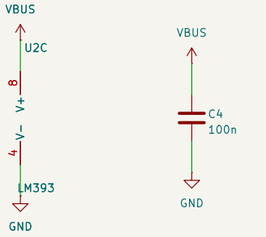


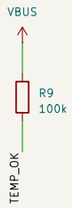
R_PROG 10k -> I_REG = 1000 V / 10 kOhm = 100 mA (MCP73831 datasheet).
PROG reaches GND only through Q1: Q1 off -> PROG floats -> charger disabled.
D1 lights while STAT is low (charging). Charge complete or inhibited = off.

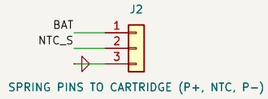
03 PACK NTC WINDOW (HARDWARE, NO FIRMWARE)

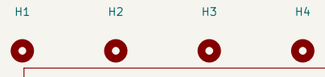












NTC_S = VBUS x R_NTC / (22k + R_NTC). Window (ratio of VBUS): cold/open trip 0.57 (VTH_HI), hot/short trip 0.18 (VTH_LO).
10k B3435 NTC: 0 C -> 28.7k -> 0.566; 45 C -> 4.85k -> 0.181. B3950: trips at -2.5 C / -43 C (conservative).
No cartridge: NTC_S = VBUS (5 V) -> unit A pulls TEMP_OK low -> no charge. Only VTH_HI (2.85 V) must be inside the LM393 common-mode range (VCC - 1.5 V at 25 C, VCC - 2 V over temperature): an input outside it still gives a correct output.
No hysteresis: charge may toggle at a threshold, which is benign (charging is inhibited outside the window).

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.
NIVA

Sheet: /
File: NIVA-dock.kicad_sch

Title: NIVA cartridge charging dock | USB-C, MCP73831, hardware NTC windo

Size: A3	Date: 2026-09-28	**Rev: C — ENGINEERING PROTO**
KiCad E.D.A. 10.0.7		Id: 1/1